



MDMC

Master in Data Management
and Curation



Git for beginners

Gianfranco Gallizia,
Trieste, October 2nd
2026



Rebase.

Recap

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Rebase.

Recap

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Best practices.

Git creates snapshots of the tracked files and calculates the SHA-1 hash of these snapshots. These hashes identify the right binary file (*blob*) inside the `GIT_DIR`.

Rebase.

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Git creates snapshots of the tracked files and calculates the SHA-1 hash of these snapshots. These hashes identify the right binary file (*blob*) inside the `GIT_DIR`.

A commit is an entry containing the metadata linked to a specific change:

Rebase.

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Git creates snapshots of the tracked files and calculates the SHA-1 hash of these snapshots. These hashes identify the right binary file (*blob*) inside the `GIT_DIR`.

A commit is an entry containing the metadata linked to a specific change:

- Author's name.
- Author's email.
- Date and time of commit's creation.
- SHA-1 of the blob.
- SHA-1 of the parent commit(s).
- SHA-1 of all the previous metadata.

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Tree

A system of chained commits from which it is possible to go back to the first commit entered, regardless of the starting point.

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Tree

A system of chained commits from which it is possible to go back to the first commit entered, regardless of the starting point.

Nota bene:

The root of the tree is **the last commit inserted** (HEAD), not the first.

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Tree

A system of chained commits from which it is possible to go back to the first commit entered, regardless of the starting point.

Nota bene:

The root of the tree is **the last commit inserted** (HEAD), not the first.

A commit creates a new root in the tree, whereas a merge combines two separate sub-trees by giving them a new shared root.

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Git's standard approach is to have developers work independently on their own sub-trees, then combine their work through merges. During a merge, any conflicting changes are reconciled and resolved in the resulting commit.

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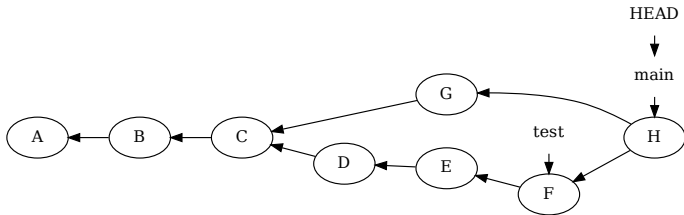
Rebase.

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Best practices.

Git's standard approach is to have developers work independently on their own sub-trees, then combine their work through merges. During a merge, any conflicting changes are reconciled and resolved in the resulting commit.



Rebase.

Definition

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Rebase.

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Rebase

Reposition the development branch so that it begins from the latest point in the main branch.

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Let's go back:

Rebase.

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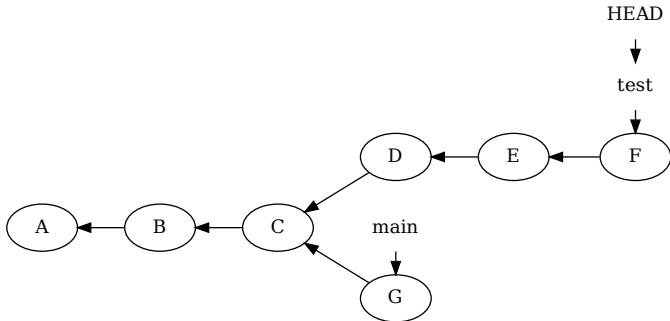
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Help!

Best practices.

Let's go back:



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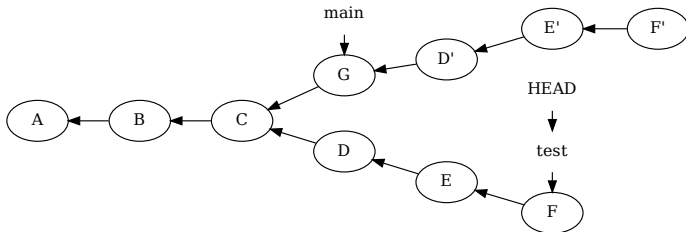
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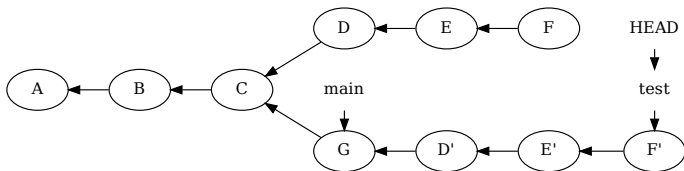
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Definition

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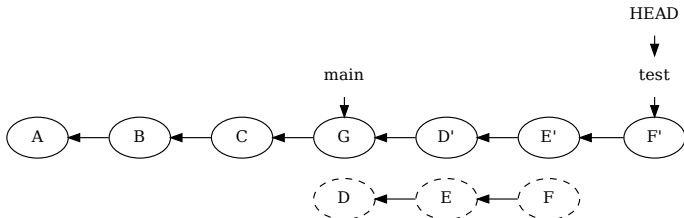
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Rebase.

Issues

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Issues

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Conflicts

In case of conflicting changes git will stop the rebase:

```
git rebase main
Auto-merging README.txt
CONFLICT (content): Merge conflict in README.txt
error: could not apply cf21f66... Lowercase.
[omissis...]
Could not apply cf21f66... Lowercase.
```

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Issues

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Conflicts

Git also provides some suggestions on how to proceed:

Resolve all conflicts manually, mark them as resolved with
"git add/rm <conflicted_files>", then run "git rebase --continue".
You can instead skip this commit: run "git rebase --skip".

To abort and get back to the state before "git rebase", run "git rebase --abort".

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Issues

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WARNING

Rebasing **rewrites the history of a branch**. This isn't a problem for a local branch that only exists on our machine, but it becomes a major issue once we've already distributed our changes to others.

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How to perform a rebase

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To perform a rebase:

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To perform a rebase:

- 1 Switch to the branch **that you want to rebase** using
`git checkout $BRANCH`

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To perform a rebase:

- 1 Switch to the branch **that you want to rebase** using
`git checkout $BRANCH`
- 2 Run the command `git rebase $BASE`, where `$BASE` is
the branch that you want to use as the new starting point.

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To perform a rebase:

- 1 Switch to the branch **that you want to rebase** using `git checkout $BRANCH`
- 2 Run the command `git rebase $BASE`, where `$BASE` is the branch that you want to use as the new starting point.
- 3 Wait for the rebase to complete or for it to be interrupted due to conflicts.

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To perform a rebase:

- 1 Switch to the branch **that you want to rebase** using
`git checkout $BRANCH`
- 2 Run the command `git rebase $BASE`, where `$BASE` is
the branch that you want to use as the new starting point.
- 3 Wait for the rebase to complete or for it to be interrupted
due to conflicts.
- 4 If the rebase is interrupted, **resolve the conflicts** and
then continue the rebase with
`git rebase --continue`

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Patches

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Patches

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My collaborators don't use git. . .

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My collaborators don't use git...

PATCH(1) General Commands Manual PATCH(1)

NAME

patch - apply a diff file to an original

SYNOPSIS

patch [options] [originalfile [patchfile]]

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Sending and receiving patches was the historical method of collaboration between developers.

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Best practices.

Sending and receiving patches was the historical method of collaboration between developers.

The main advantage of this method is that you are not tied to a specific Version Control System. The resulting files can be read, sent to the `patch` command and integrated without the need for the receiver to install git.

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Best practices.

Sending and receiving patches was the historical method of collaboration between developers.

The main advantage of this method is that you are not tied to a specific Version Control System. The resulting files can be read, sent to the `patch` command and integrated without the need for the receiver to install git.

The main disadvantage is that it's an obsolete way of collaboration without authentication and it's based on mutual trust between people. Furthermore those receiving the patches must do some work to integrate those changes to their codebase.

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SSH

Network protocol at the Application level of the TCP/IP stack usable by git to send data between hosts.

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Best practices.

SSH

Network protocol at the Application level of the TCP/IP stack usable by git to send data between hosts.

Big emphasis on security: through SSH all the communications are authenticated (both the user and the machines are checked) and performed via encrypted tunnels over the network according to a set of algorithms negotiated between the hosts.

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Best practices.

The basic idea

Create a repository on an host with SSH active, instruct the clients to send their changes through SSH to said host and to retrieve our changes though SSH from that host.

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Best practices.

The basic idea

Create a repository on an host with SSH active, instruct the clients to send their changes through SSH to said host and to retrieve our changes though SSH from that host.

Nota bene:

The host can be any machine and theoretically all the developers could open the access to their machines through SSH and send data between them in a peer-to-peer network.

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Server configuration

Sharing your work.

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Server configuration

We will not do it, if you are interested there's a whole chapter dedicated to the server setup for git in the Git Book (<https://git-scm.com/book/en/v2/>)

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Client configuration

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Client configuration

If you want to collaborate to the development of a project that already uses git through SSH follow these steps:

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Client configuration

If you want to collaborate to the development of a project that already uses git through SSH follow these steps:

- 1 Ask for an account; or register with the organization/git service provider and then ask for access.

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Best practices.

Client configuration

If you want to collaborate to the development of a project that already uses git through SSH follow these steps:

- 1 Ask for an account; or register with the organization/git service provider and then ask for access.
- 2 Obtain the repository URL in the format
`ssh://user@git.example.com/org/project.git`

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Client configuration

If you want to collaborate to the development of a project that already uses git through SSH follow these steps:

- 1 Ask for an account; or register with the organization/git service provider and then ask for access.
- 2 Obtain the repository URL in the format
`ssh://user@git.example.com/org/project.git`
- 3 Use the `git clone $URL` command (where \$URL is the one previously obtained).

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Best practices.

Client configuration (cont.)

`git clone` will connect through SSH to `git.example.com` and then start downloading the repository in a directory called `project`. Once finished you will have a local copy of the whole repository and your working copy will be automatically placed on the last commit of the default branch.

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Git through SSH - git clone

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Cloning in a specific directory:

```
git clone $URL $WORKDIR
```

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Best practices.

Cloning in a specific directory:

```
git clone $URL $WORKDIR
```

Cloning just the last commit:

```
git clone --depth 1 $URL
```

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Git through SSH - git clone

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Best practices.

Cloning in a specific directory:

```
git clone $URL $WORKDIR
```

Cloning just the last commit:

```
git clone --depth 1 $URL
```

Cloning just the last commit of a specific branch:

```
git clone --single-branch -b $BRANCH --depth 1 $URL
```

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Best practices.

Cloning a remote repository keeping separate GIT_DIR and
WORKDIR:

```
git clone --separate-git-dir=$GIT_DIR $URL $WORKDIR
```

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Best practices.

Cloning a remote repository keeping separate GIT_DIR and WORKDIR:

```
git clone --separate-git-dir=$GIT_DIR $URL $WORKDIR
```

Cloning only GIT_DIR get a *bare repository*:

```
git clone --bare $URL $GIT_DIR
```

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Best practices.

Remote

In git a *remote* is an URL pointing to a remote repository. Git can have more than one remote per local repository.

For each local repository there can only be one *upstream* remote. The *upstream* remote is the default remote for the data transfer operations.

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Connecting to a remote

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Connecting to a remote

1 Add a remote.

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Connecting to a remote

- 1 Add a remote.
- 2 Send your changes.

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Connecting to a remote

- 1 Add a remote.
- 2 Send your changes.
- 3 (Optional) send your tags.

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Best practices.

Connecting to a remote

- 1 Add a remote.
- 2 Send your changes.
- 3 (Optional) send your tags.

```
git remote add $REMOTE_NAME $URL
git push --all --set-upstream $REMOTE_NAME
git push --tags
```

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Interlude - tags

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Best practices.

Tags

You can assign a human readable pointer to a commit (*tag a commit*). Usually this operation is performed by the maintainer to indicate which commit has been used to create a specific version of the software.

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Interlude - tags

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Best practices.

Tags

You can assign a human readable pointer to a commit (*tag a commit*). Usually this operation is performed by the maintainer to indicate which commit has been used to create a specific version of the software.

Assigning a tag:

```
git tag $LABEL $COMMIT
```

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Interlude - specify a commit

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Best practices.

Ways to specify a commit:

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Ways to specify a commit:

Full hash: 284a5a91c43d7a2e0ceb1eb4721dca08da17b3a0

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Interlude - specify a commit

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Ways to specify a commit:

Full hash: 284a5a91c43d7a2e0ceb1eb4721dca08da17b3a0

Shortened hash: 284a5a91

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Ways to specify a commit:

Full hash: 284a5a91c43d7a2e0ceb1eb4721dca08da17b3a0

Shortened hash: 284a5a91

Last commit: HEAD

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Interlude - specify a commit

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Ways to specify a commit:

Full hash: 284a5a91c43d7a2e0ceb1eb4721dca08da17b3a0

Shortened hash: 284a5a91

Last commit: HEAD

Branch name: `main`

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Interlude - specify a commit

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Best practices.

Ways to specify a commit:

Full hash: 284a5a91c43d7a2e0ceb1eb4721dca08da17b3a0

Shortened hash: 284a5a91

Last commit: HEAD

Branch name: main

Relative (3 commits before main): main^3 or main~3

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Retrieve other people changes

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Retrieve other people changes

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Retrieve other people changes

First we make sure we are ready to retrieve the changes from the remote repository. To do so we will use the `git status` command:

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Retrieve other people changes

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Best practices.

Retrieve other people changes

First we make sure we are ready to retrieve the changes from the remote repository. To do so we will use the `git status` command:

```
git status
```

```
On branch main
```

```
Your branch is up to date with 'origin/main'.
```

```
nothing to commit, working tree clean
```

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Retrieve other people changes

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Retrieve from upstream remote (default):

```
git pull
```

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Retrieve other people changes

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Retrieve from upstream remote (default):

```
git pull
```

Retrieve from a different remote:

```
git pull $REMOTE_NAME $REMOTE_BRANCH
```

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Interlude - pull strategy

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Interlude - pull strategy

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Pull strategy

Unless otherwise specified; `git pull` will try to *merge* the remote branch into the local branch.

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Pull strategy

Unless otherwise specified; `git pull` will try to *merge* the remote branch into the local branch.

When possible git will perform a *fast forward*, otherwise it will create a *merge commit*.

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Pull strategy

Unless otherwise specified; `git pull` will try to *merge* the remote branch into the local branch.

When possible git will perform a *fast forward*, otherwise it will create a *merge commit*.

If you want to apply your local changes on top of the remote changes you can opt for the *rebase* strategy:

```
git pull --rebase
```

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Pull strategy

Unless otherwise specified; `git pull` will try to *merge* the remote branch into the local branch.

When possible git will perform a *fast forward*, otherwise it will create a *merge commit*.

If you want to apply your local changes on top of the remote changes you can opt for the *rebase* strategy:

```
git pull --rebase
```

Nota bene:

Everything we said regarding merge, conflicts and rebase apply here.

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GitHub Workflow

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GitHub Workflow

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Collaborations

Each project has its own rules and it's hard to give some recommendations that are generic enough to be applicable everywhere. Except for the very basic ones.

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GitHub Workflow

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Best practices.

Collaborations

Each project has its own rules and it's hard to give some recommendations that are generic enough to be applicable everywhere. Except for the very basic ones.

GitHub

GitHub is the largest git service provider in the World. Each user registered on GitHub could create a repository on GitHub's servers and access through SSH or HTTPS.

Sharing your work.

GitHub Workflow

Git for
beginners

Gianfranco
Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

GitHub Workflow

Sharing your work.

GitHub Workflow

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Rebase.

Sharing your
work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.

Sharing your work.

GitHub Workflow

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Sharing your
work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.
- 2 Press the "fork" button to create a copy of the project's repository under your account.

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Sharing your
work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.
- 2 Press the "fork" button to create a copy of the project's repository under your account.
- 3 Clone locally your *fork*.

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work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.
- 2 Press the "fork" button to create a copy of the project's repository under your account.
- 3 Clone locally your *fork*.
- 4 Make your changes, create your branches and work to the project just like it was yours.

Sharing your work.

GitHub Workflow

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work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.
- 2 Press the "fork" button to create a copy of the project's repository under your account.
- 3 Clone locally your *fork*.
- 4 Make your changes, create your branches and work to the project just like it was yours.
- 5 When you're satisfied with your work push it on GitHub.

Sharing your work.

GitHub Workflow

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work.

Help!

Best practices.

GitHub Workflow

- 1 Find the project's page on GitHub.
- 2 Press the "fork" button to create a copy of the project's repository under your account.
- 3 Clone locally your *fork*.
- 4 Make your changes, create your branches and work to the project just like it was yours.
- 5 When you're satisfied with your work push it on GitHub.
- 6 Open a "Pull Request" to the project's maintainers and ask them to integrate your changes to the main project.

Sharing your work.

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Help!

Best practices.

Sharing your work.

GitHub Workflow

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Rebase.

Sharing your
work.

Help!

Best practices.

Pro

GitHub allows to create an unlimited number of public and private repositories provided that you respect the Terms and Conditions set by Microsoft.

Sharing your work.

GitHub Workflow

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Rebase.

Sharing your
work.

Help!

Best practices.

Pro

GitHub allows to create an unlimited number of public and private repositories provided that you respect the Terms and Conditions set by Microsoft.

Con

Almost all the data flows through GitHub servers: all the communications pass by GitHub web interface and all the code is shared via GitHub (regardless if it's integrated to the main project or not).

Help!

I made a mess!

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help!

I made a mess!

Git for
beginners

Gianfranco
Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

Help! I made a mess!

Help!

I made a mess!

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help! I made a mess!

1 Don't panic.

Help!

I made a mess!

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beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help! I made a mess!

- 1 Don't panic.
- 2 If you have not issued `git push` you can still fix it (usually).

Help!

I made a mess!

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help! I made a mess!

- 1 Don't panic.
- 2 If you have not issued `git push` you can still fix it (usually).
- 3 Everything that is put in `.gitignore` is under our responsibility (if we delete it: it's lost).

Help!

I made a mess!

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help! I made a mess!

- 1 Don't panic.
- 2 If you have not issued `git push` you can still fix it (usually).
- 3 Everything that is put in `.gitignore` is under our responsibility (if we delete it: it's lost).
- 4 Everything that is not tracked by git is under our responsibility (if we delete it: it's lost).

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I wrote the wrong commit message!

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I wrote the wrong commit message!

```
git commit --amend
```

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've put the wrong content inside the commit!

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've put the wrong content inside the commit!

```
git reset --soft HEAD^1
# Edit the file with a text editor
git add -i
git commit
```

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've committed to the wrong branch!

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've committed to the wrong branch!

```
git log --oneline
git checkout dev
git cherry-pick $HASH_OF_THE_LAST_MAIN_COMMIT
git checkout main
git reset --hard HEAD~1
```

Help!

Amend a commit

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've committed to the wrong branch!

```
git log --oneline
git checkout dev
git cherry-pick $HASH_OF_THE_LAST_MAIN_COMMIT
git checkout main
git reset --hard HEAD^1
```

IMPORTANT!

`git reset --soft HEAD^1` deletes the last commit of the current branch and leaves the working copy unaltered. `git reset --hard HEAD^1` deletes the last commit of the current branch **AND** the relative changes from working copy.

Help!

A complex example

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Rebase.

Sharing your
work.

Help!

Best practices.

Help!

A complex example

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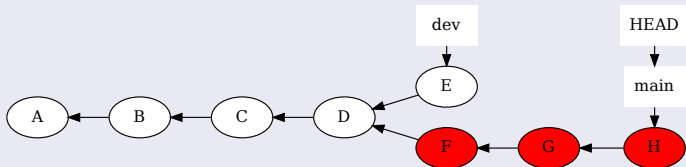
Rebase.

Sharing your
work.

Help!

Best practices.

I made a bunch of commits on the wrong branch!



Help!

A complex example

Git for
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Rebase.

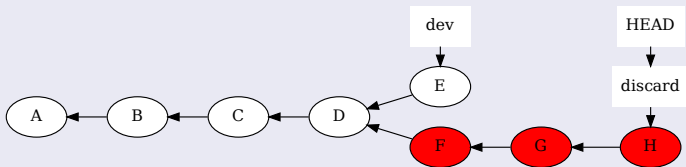
Sharing your
work.

Help!

Best practices.

Rename main in discard

```
git branch -m discard
```



Help!

A complex example

Git for
beginners

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Gallizia

Rebase.

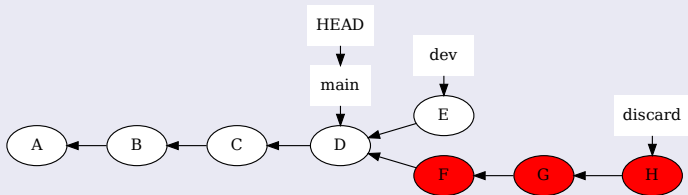
Sharing your
work.

Help!

Best practices.

Create a new branch called `main` ending in D

```
git checkout $D  
git branch main
```



Help!

A complex example

Git for
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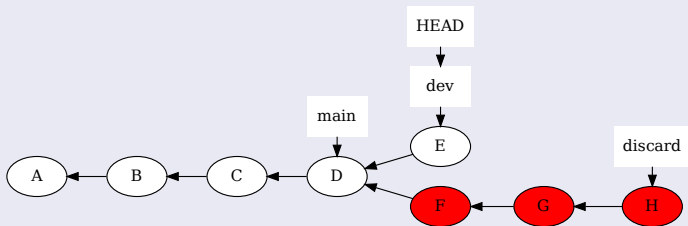
Rebase.

Sharing your
work.

Help!

Best practices.

Let's move on dev
`git checkout dev`



Aiuto!

Un esempio complesso

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Rebase.

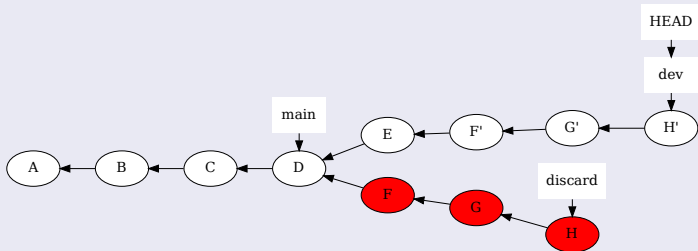
Sharing your
work.

Help!

Best practices.

Let's copy F, G and H in dev

```
git cherry-pick main..discard
```



Help!

A complex example

Git for
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Gallizia

Rebase.

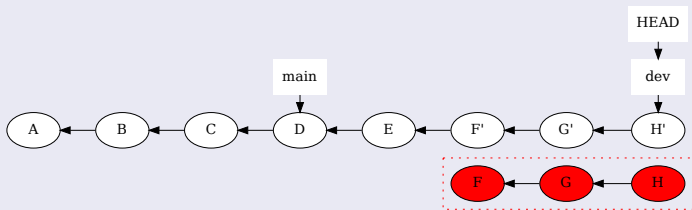
Sharing your
work.

Help!

Best practices.

Delete the discard branch

```
git branch -D discard
```



Help!

Pull rebase

Git for
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Rebase.

Sharing your
work.

Help!

Best practices.

Help!

Pull rebase

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've committed on the default branch before pulling the remote changes!

Help!

Pull rebase

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I've committed on the default branch before pulling the remote changes!

```
git pull --rebase
```

Help!

After a push

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I pushed something I shouldn't have pushed...

Help!

After a push

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

I pushed something I shouldn't have pushed...

I'm sorry. Create a new commit where you fix everything, write "Revert wrong modifications." as the commit message, issue `git push` and embrace the shame.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Best practices.

Things to do

Git for
beginners

Gianfranco
Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

git status mantra

Best practices.

Things to do

Git for
beginners

Gianfranco
Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

git status mantra

Force yourself to issue `git status` **before and after** issuing another git command.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

git status mantra

Force yourself to issue `git status` **before and after** issuing another git command.

`git status` shows a snapshot of your working copy, which files you have placed in the staging area to be committed, and also suggests what to do during merge and rebase operations.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Local branches are free

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Local branches are free

If you're unsure about issuing `git checkout -b new_branch` before moving on: create the branch.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Local branches are free

If you're unsure about issuing `git checkout -b new_branch` before moving on: create the branch.

Branching in other VCSs is an epic event reserved to a few blessed individuals, branching in git is something that's done multiple times per day sometimes.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

1 Check out a local branch.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

- 1 Check out a local branch.
- 2 Make your changes.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

- 1 Check out a local branch.
- 2 Make your changes.
- 3 Go back to the starting branch.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

- 1 Check out a local branch.
- 2 Make your changes.
- 3 Go back to the starting branch.
- 4 `git pull`

Best practices.

Things to do

Git for
beginners

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Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

- 1 Check out a local branch.
- 2 Make your changes.
- 3 Go back to the starting branch.
- 4 `git pull`
- 5 `git log --all --decorate --oneline --graph`

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

pull comes always before push

- 1 Check out a local branch.
- 2 Make your changes.
- 3 Go back to the starting branch.
- 4 `git pull`
- 5 `git log --all --decorate --oneline --graph`
- 6 Decide if you want to merge the local branch, rebase the local branch or `git cherry-pick`, etc. etc.

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Bonus: be careful with aliases

Best practices.

Things to do

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Bonus: be careful with aliases

```
git config --global alias.adog "log --all \
--decorate --oneline --graph"
git adog
* 8b393c8 (HEAD -> change_B) Merge branch\ldots
|\
| * 5855fe7 (change_A) Lowercase.
* | bd1130e (main) Double the letters.
|/
* d1f0f78 Initial Commit.
```

Best practices.

Things to avoid

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Best practices.

Things to avoid

Git for
beginners

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Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

```
git commit -a -m '.'
```

Best practices.

Things to avoid

Git for
beginners

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Gallizia

Rebase.

Sharing your
work.

Help!

Best practices.

```
git commit -a -m '.'
```

"I don't care, just save it."

Best practices.

Things to avoid

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beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

```
git push --force
```

Best practices.

Things to avoid

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

```
git push --force
```

Git refuses to complete the push action if it detects some unreconcilable differences between the local repository and the remote repository.

Best practices.

Things to avoid

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

```
git push --force
```

Git refuses to complete the push action if it detects some unreconcilable differences between the local repository and the remote repository.

`git push --force` tells git to continue with the operation and to **overwrite** the changes with those present in the local branch. Basically you're being a dictator and you are implicitly telling your collaborators that their work is not as important as yours.

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Best practices.

Use case - LADE

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

Best practices.

Use case - LADE

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beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.

Best practices.

Use case - LADE

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.
- The person responsible evaluates the merge request and can do one of the following actions:

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.
- The person responsible evaluates the merge request and can do one of the following actions:
 - Accept the merge request as is.

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.
- The person responsible evaluates the merge request and can do one of the following actions:
 - Accept the merge request as is.
 - Give some suggestions in the form of comments to the merge request.

Best practices.

Use case - LADE

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beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.
- The person responsible evaluates the merge request and can do one of the following actions:
 - Accept the merge request as is.
 - Give some suggestions in the form of comments to the merge request.
 - Modify the merge request by pushing their changes to the merge request branch.

Best practices.

Use case - LADE

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Rebase.

Sharing your
work.

Help!

Best practices.

How Git is used in LADE

- Each repository has a person responsible for that repository.
- Each member of LADE can create additional branches.
- When their changes are ready they open a **merge request** on Gitlab.
- The person responsible evaluates the merge request and can do one of the following actions:
 - Accept the merge request as is.
 - Give some suggestions in the form of comments to the merge request.
 - Modify the merge request by pushing their changes to the merge request branch.
 - Refuse the merge request and close it.

Best practices.

Use case - LADE

Git for
beginners

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Rebase.

Sharing your
work.

Help!

Best practices.

Pro

Best practices.

Use case - LADE

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Rebase.

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Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.

Best practices.

Use case - LADE

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Rebase.

Sharing your
work.

Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.

Best practices.

Use case - LADE

Git for
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Rebase.

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work.

Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.
- There are no doubts on *what* is the repository to clone.

Best practices.

Use case - LADE

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Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.
- There are no doubts on *what* is the repository to clone.

Con

Best practices.

Use case - LADE

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Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.
- There are no doubts on *what* is the repository to clone.

Con

- The person responsible for the repository must resolve the conflicts or convince the proposer to do it.

Best practices.

Use case - LADE

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Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.
- There are no doubts on *what* is the repository to clone.

Con

- The person responsible for the repository must resolve the conflicts or convince the proposer to do it.
- History is not linear.

Best practices.

Use case - LADE

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work.

Help!

Best practices.

Pro

- People don't get stuck waiting for their changes to be approved.
- There's a check on what gets into the main branch.
- There are no doubts on *what* is the repository to clone.

Con

- The person responsible for the repository must resolve the conflicts or convince the proposer to do it.
- History is not linear.
- If I have an open merge request and my changes are not part of the main branch I must merge from the main branch, I cannot rebase.

End of module 2

Git for
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Rebase.

Sharing your
work.

Help!

Best practices.

Thank you for your attention!

End of module 2

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Rebase.

Sharing your
work.

Help!

Best practices.

Thank you for your attention!

Any questions?



OGS

Istituto Nazionale
di Oceanografia
e di Geofisica
Sperimentale

ARCTUR



REGIONE AUTONOMA
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E-ARGO